

This may seem like an odd format for a proposal, but it's structured to assist you in making sure your proposal meets the project requirements. To get started, answer the questions below. Then, you'll be taking your responses and using them to craft a proposal/synopsis (last page) and finally, set some deadlines for yourself in a timeline.

1. What is the new statistical topic you will be exploring for the project? If you have a large topic area in mind (like "clustering" or "imputation"), you should work (with me!) to narrow it down to one or several sub-topics (methods) within the area.

I will be exploring the k-nearest neighbors (kNN) algorithm, focusing primarily on its use in classification problems.

2. Consider where you have encountered this topic. Why do you consider this topic "new" to you? What background do you already have in relation to the topic? (This part isn't included in the proposal/synopsis below, but I need to know your background.)

Although kNN has been briefly mentioned in two of my previous classes (STAT-495 and COSC-247), it was only described at a very high level, essentially, "a method that predicts outcomes based on nearby data points." Beyond that, I have not studied the algorithm in depth, so the underlying concepts, implementation, and evaluation are all new to me.

3. The audience for the paper is a peer in the class, who won't necessarily have had the same electives/background experiences as you. Besides details about the new statistical topic, what other topics will your peers need information about in order to understand the topic? (For example, if you want to study a particular method of imputation, you are going to need to describe the types of missing data and imputation methods in general in order for it to be understandable to a classmate). If you don't know much about your topic, you may not be able to fill in much about this yet.

I haven't thought about this in detail yet, but I think I will begin with an overview of supervised learning. Specifically, what classification and regression algorithms aim to accomplish. Then I will introduce key ideas behind distance-based prediction so the motivation for kNN is clear.

4. Are you currently leaning more towards an application or a simulation study for the required second component? Provide some reasoning about why this is your current thinking. You may or may not have very specific ideas here yet.

I am leaning more towards an application at this point. I think kNN is an intuitive method to understand visually, and seeing it applied to a real dataset will help highlight how proximity-based prediction works. Also I don't know how I would set up a simulation study in this context just yet.

5. If you have any other thoughts on planned applications (such as specific topic areas you want it to be in) or things you'd want to do in your simulations, feel free to list those here.

I would like to do an application in tech app recommendations i.e. apply on a kNN dataset to see what recommendations would be given to a user based on their past activities I recognize that finding the right dataset may be challenging, but I will explore available sources. I also plan to compare kNN's performance with at least one other algorithm.

6. The first major project component is an expository review / exposition (an extremely solid Literature / Background / Introduction to the topic section, whatever name you give it) of the new topic. This is likely to include additional information about the general statistical topic area, and example applications in the literature. Thinking of your statistical and writing skills (and goals you have set for class), what do you expect will be the most challenging aspect of preparing this exposition for you? (Expositions may have a code component too and include toy examples.) What skills will this section help you demonstrate?

I think coming up with toy examples seems like will be the most challenging part for me. Because of project 1 and 2, I feel comfortable with the writing aspect of it. This section will help me demonstrate my ability to communicate statistical concepts to an audience who may be unfamiliar with the topic.

7. The application / simulation study (or both, etc.) that you include will mean some coding (in R or another software). What do you expect will be the most challenging aspect of preparing this section and reporting the findings from your work? (This includes coding, statistical, and writing skills). What skills will this section help you demonstrate?

Using new R packages and configuring kNN properly will likely be the biggest challenge. This will help me demonstrate coding skills, model evaluation and clear reporting of results.

Now, let's synthesize all of this into a complete proposal via making it a project synopsis – e.g. you lay out what you plan to do. Here is an example assuming that random forests were the topic (this is not “new” for any of you though!), so I can demonstrate how these responses (from the previous pages) go together. See if you can see how each sentence relates to a question above. Space for you to write your own synopsis (which is what I'll focus on reading in this submission) is below the example.

Example Proposed Project Title: Using Random Forests to Predict Company Bankruptcy

Example Project Synopsis:

For my project, I plan to explore the use of random forests in supervised learning (classification) problems. My paper will introduce my peers to the classification setting, review the concept of decision trees in order to motivate forests, and then delve into the details of random forests, as well as touch on the pre-cursor of bagging. I will demonstrate how random forests work with a toy example such as predicting Species in iris in the exposition to help make points clear for my peers. I will also include several examples of applications of random forests in the literature to demonstrate recent use of the method. For my application, I intend to apply random forests to a larger data set (potentially sourced from Kaggle) with a binary response that indicates whether a company is bankrupt or not. I intend to

use a new R package for fitting the random forest, called ranger. Wrangling the data set will let me demonstrate appropriate wrangling skills and reporting my results will show my ability to communicate my data analysis process. I will include a comparison to logistic regression to showcase my modeling and variable selection skills in a regression setting, before comparing the models based on performance.

Note: In the synopsis, I tried to write about intentions (if some things were uncertain) or write about what I “will” do. This helps provide a road map for you – you know you need to do those things. Your synopsis may not have some of these details, but I am hoping writing it this way helps you envision what the final product is and how you can work to get there. Sharing intentions is useful! You can write that you intend to find a data set that allows you to demonstrate skills X, Y, and Z, or that you intend to write a simulation to demonstrate A and B, etc.

Your turn

Your name: Daizy Buluma

Proposed Project Title: Exploring k-Nearest Neighbors

Project Synopsis:

For my project, I plan to explore the k-nearest neighbors (kNN) method in supervised learning. My paper will begin by introducing the classification and regression framework to provide context for where kNN is used and then explain how kNN makes predictions based on similarity to nearby observations. I will include a simple toy example to help demonstrate how distance plays a role in classifying new data points.

I also plan to incorporate examples of kNN applications found in published work to illustrate its relevance in real settings. For the applied component of my project, I aim to analyze a dataset related to tech app recommendations or user behavior, using kNN to predict what apps a user might be interested in based on similar users. I intend to compare the performance of kNN with at least one additional supervised learning model to evaluate whether kNN is the best method for this type of prediction task.

Through this project, I will develop my understanding of kNN and practice communicating complex concepts clearly to peers.

Timeline

Finally, we want to consider a timeline. The deliverable deadlines are listed here to assist you. Remember that the deliverables are focused around the “new” part of the project – the exposition piece. You should make sure to include deadlines for yourself for the application / simulation study piece as well, since you need to balance them. Outline some dates and associated deadlines for yourself below (or interspersed) with the list below.

Wednesday, Nov 5 — Project Proposal due

Friday, Nov 7 — Begin literature review and search for foundational sources on kNN

Monday, Nov 10 — Continue literature review and note-taking

Wednesday, Nov 12 — Finalize annotated bibliography

Thursday, Nov 13 — Annotated Bibliography Due

Friday, Nov 14 — Draft exposition outline

Monday, Nov 17 — Write exposition sections: Intro to classification vs regression; What is kNN?

Wednesday, Nov 19 — Write exposition sections: Technical details, tuning k, normalization, example use cases

Friday, Nov 21 — Exposition Draft Due

Thanksgiving break - find/analyze dataset, test simple kNN model, gather figures (if possible)

Monday, Dec 1 — Select final dataset; begin data wrangling

Wednesday, Dec 3 — Fit kNN model and baseline model; run performance metrics

Friday, Dec 5 — Draft for Peer Review Due (target: core modeling + results completed)

Monday, Dec 8 — Peer Review Discussion (in class)

Wednesday, Dec 10 — Check-in Due

Friday, Dec 12 — Final polishing, visuals, proofreading

Monday, Dec 15 — Final Report Due

Dec 17 or 18 — Reflection Discussion